

$$21. H = -\hbar\Delta |e\rangle\langle e| + \frac{\hbar\Omega}{2} \sigma_x$$

$$\dot{\sigma}_- (t) = \frac{i}{\hbar} [H, \sigma_-] - \underbrace{[\sigma_-, \sigma_+]}_{\sigma_z} \left(\frac{\gamma}{2} \sigma_- + \sqrt{\gamma} b_{in} \right) - \underbrace{\left(\frac{\gamma}{2} \sigma_+ + \sqrt{\gamma} b_{in}^\dagger \right)}_{=0} [\sigma_-, \sigma_-] \quad \text{decay} \quad t := t(t)$$

$$- \underbrace{[\sigma_-, \sigma_z]}_{-2\sigma_-} \left(\frac{\gamma}{2} \sigma_z + \sqrt{\gamma} b_{in} \right) - \underbrace{\left(\frac{\gamma}{2} \sigma_z + \sqrt{\gamma} b_{in}^\dagger \right)}_{=0} [\sigma_z, \sigma_-] \quad \text{dephasing}$$

since everything is evaluated at time t the b_{in} cancel or are resorted, as everything is denoted dependent on time t

$$= \frac{i}{\hbar} \left(\underbrace{[-\hbar\Delta |e\rangle\langle e|, \sigma_-]}_{-\sigma_-} + \underbrace{\frac{\hbar\Omega}{2} [\sigma_x, \sigma_-]}_{-\sigma_z} \right) - \sigma_z \left(\frac{\gamma}{2} \sigma_- + \sqrt{\gamma} b_{in} \right) - (-2\sigma_-) \left(\frac{\gamma}{2} \sigma_z + \sqrt{\gamma} b_{in} \right) - \left(\frac{\gamma}{2} \sigma_z + \sqrt{\gamma} b_{in}^\dagger \right) (2\sigma_-)$$

$$= i\Delta\sigma_- - i\frac{\Omega}{2}\sigma_z - \frac{\gamma}{2}\sigma_- - \sqrt{\gamma}\sigma_z b_{in} - \gamma\sigma_- + \sqrt{\gamma}2\sigma_- b_{in} - \gamma\phi\sigma_- - 2\sqrt{\gamma}b_{in}^\dagger\sigma_-$$

$$= (i\Delta - \frac{\gamma}{2} - \gamma\phi - 2\sqrt{\gamma}b_{in}^\dagger)\sigma_- + 2\sqrt{\gamma}\sigma_- b_{in} + \sigma_z(-i\frac{\Omega}{2} - \sqrt{\gamma}b_{in})$$

$$(\sigma_z b_{in})^\dagger = b_{in}^\dagger \sigma_z^\dagger = b_{in}^\dagger \sigma_z$$

$$\dot{\sigma}_+ (t) = (\sigma_-^\dagger (t)) = -i\Delta\sigma_+ + i\frac{\Omega}{2}\sigma_z - \frac{\gamma}{2}\sigma_+ - \sqrt{\gamma}b_{in}^\dagger\sigma_z - \gamma\phi\sigma_+ + 2\sqrt{\gamma}b_{in}^\dagger\sigma_+ - \gamma\phi\sigma_+ - 2\sqrt{\gamma}\sigma_+ b_{in}$$

$$= (-i\Delta - \frac{\gamma}{2} - \gamma\phi + 2\sqrt{\gamma}b_{in}^\dagger)\sigma_+ - 2\sqrt{\gamma}\sigma_+ b_{in} + (i\frac{\Omega}{2} - \sqrt{\gamma}b_{in}^\dagger)\sigma_z$$

$$\dot{\sigma}_z (t) = ([\sigma_-, \sigma_+](t)) = (\dot{\sigma}_- \sigma_+ - \sigma_+ \dot{\sigma}_-)$$

$$= \dot{\sigma}_- \sigma_+ + \sigma_- \dot{\sigma}_+ - \dot{\sigma}_+ \sigma_- - \sigma_+ \dot{\sigma}_-$$

$$\sigma_- \sigma_+ = |g\rangle\langle g|$$

$$\sigma_z \sigma_+ = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \sigma_+$$

$$\sigma_- \sigma_z = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} = -\sigma_-$$

$$\begin{aligned} &= \left((i\Delta - \frac{\gamma}{2} - \gamma\phi - 2\sqrt{\gamma}b_{in}^\dagger) |g\rangle\langle g| + 2\sqrt{\gamma}\sigma_- b_{in} \sigma_+ + i\frac{\Omega}{2}\sigma_+ - \sqrt{\gamma}\sigma_z b_{in} \sigma_+ \right. \\ &+ |g\rangle\langle g| (-i\Delta - \frac{\gamma}{2} - \gamma\phi - 2\sqrt{\gamma}\phi b_{in}) + 2\sqrt{\gamma}\sigma_- b_{in}^\dagger \sigma_+ - i\frac{\Omega}{2}\sigma_- - \sqrt{\gamma}\sigma_- b_{in}^\dagger \sigma_z \\ &- [(-i\Delta - \frac{\gamma}{2} - \gamma\phi + 2\sqrt{\gamma}b_{in}^\dagger) |e\rangle\langle e| - 2\sqrt{\gamma}\sigma_+ b_{in} \sigma_- + (i\frac{\Omega}{2} - \sqrt{\gamma}b_{in}^\dagger)\sigma_z] \\ &- [|e\rangle\langle e| (i\Delta - \frac{\gamma}{2} - \gamma\phi + 2\sqrt{\gamma}\phi b_{in}) - 2\sqrt{\gamma}\sigma_+ b_{in} \sigma_- + \sigma_+ (-i\frac{\Omega}{2} - \sqrt{\gamma}b_{in}^\dagger)] \end{aligned}$$

to evaluate these the systems init state is needed

b) all $\gamma\phi$ contributions should cancel hm.

$$\langle \dot{\sigma}_- \rangle = (i\Delta - \frac{\gamma}{2} - \gamma\phi) \langle \sigma_- \rangle - i\frac{\Omega}{2} \langle \sigma_z \rangle$$

both in ground state

$$= (i\Delta - \Gamma_1) \langle \sigma_- \rangle - i\frac{\Omega}{2} \langle \sigma_z \rangle$$

compare to 26.05.2025

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→ not quite right?

$$\langle \dot{\sigma}_+ \rangle = (-i\Delta - \Gamma_1) \langle \sigma_+ \rangle + i\frac{\Omega}{2} \langle \sigma_z \rangle$$

$$\langle \dot{\sigma}_z \rangle = -i\Omega \langle \sigma_- \rangle + i\Omega \langle \sigma_+ \rangle - \gamma \langle \sigma_z \rangle + \text{terms}(b_{in}, b_{in}^\dagger)$$

$$= i\Omega \langle \sigma_- \rangle - i\Omega \langle \sigma_+ \rangle - \gamma \langle \sigma_z \rangle - \gamma$$

where does my mistake come from?!

where does it come from?

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} = 0$$

$$c) \partial_t (\sigma_+(t) \sigma_-(t)) = \dot{\sigma}_+ \sigma_- + \sigma_+ \dot{\sigma}_- \stackrel{?}{=} \partial_t \sigma_{ee}(t)$$

$$\dot{\sigma}_{ee}(t) = \frac{i}{\hbar} [H, \sigma_{ee}] - \underbrace{[\sigma_{ee}, \sigma_+]}_{\sigma_+} \left(\frac{\gamma}{2} \sigma_- + \sqrt{\gamma} b_{in} \right) - \underbrace{\left(\frac{\gamma}{2} \sigma_+ + \sqrt{\gamma} b_{in}^\dagger \right)}_{\sigma_-} [\sigma_-, \sigma_{ee}] - \underbrace{[\sigma_{ee}, \sigma_z]}_{\sigma_z} \left(\frac{\gamma}{2} \sigma_z + \sqrt{\gamma} b_{in} \right) - \underbrace{\left(\frac{\gamma}{2} \sigma_z + \sqrt{\gamma} b_{in}^\dagger \right)}_{\sigma_z} [\sigma_z, \sigma_{ee}]$$

$$= i\frac{\Omega}{2} [\sigma_x, \sigma_{ee}] - \frac{\gamma}{2} \sigma_{ee} - \sqrt{\gamma} \sigma_+ b_{in} - \frac{\gamma}{2} \sigma_{ee} + \sqrt{\gamma} b_{in}^\dagger \sigma_-$$

→ equal to step in calc. of $\dot{\sigma}_z(t)$ problem above as this solution is in agreement with the OBEs? except for swapped sign on Ω terms

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \sigma_- - \sigma_+$$

b) different attempt:

rest is zero as $[\sigma_z, \sigma_z] = 0$

$$\dot{\sigma}_z = -i\Delta \underbrace{[\sigma_{ee}, \sigma_z]}_{=0} + \frac{i}{2}\Omega \underbrace{[\sigma_x, \sigma_z]}_{-2i\sigma_y} - \underbrace{[\sigma_z, \sigma_+]}_{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = -2\sigma_+} \left(\frac{\gamma}{2}\sigma_- + \sqrt{\gamma}b_{in} \right) - \left(\frac{\gamma}{2}\sigma_+ + \sqrt{\gamma}b_{in}^\dagger \right) \underbrace{[\sigma_-, \sigma_z]}_{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = -2\sigma_-}$$

$$= \Omega\sigma_y + 2\sigma_+ \left(\frac{\gamma}{2}\sigma_- + \sqrt{\gamma}b_{in} \right) - \left(\frac{\gamma}{2}\sigma_+ + \sqrt{\gamma}b_{in}^\dagger \right) (-2\sigma_-)$$

$$\sigma_y = i\sigma_+ - i\sigma_- \quad \sigma_+\sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \sigma_{ee}$$

$$\langle \dot{\sigma}_z \rangle = i\Omega\sigma_+ - i\Omega\sigma_- + \gamma\sigma_{ee} + \gamma\sigma_{ee}$$

$$= i\Omega\sigma_+ - i\Omega\sigma_- + 2\gamma \underbrace{\left(\frac{1}{2}\sigma_z + 1 \right)}_{\gamma\sigma_z + \gamma 1}$$

→ use this for OBEs

22.

Quantum Langevin Equation (QLE):

$$\dot{O}_S = \frac{i}{\hbar} [H_S, O_S] - [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{in}(t) \right) - \left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{in}^\dagger(t) \right) [c, O_S]$$

 $\sqrt{\frac{\gamma_1}{2\pi}}$

$$\dot{a}_1 = i\omega_0 \underbrace{[a_1^\dagger a_1, a_1]}_{-a_1} - [a_1, a_1^\dagger] \left(\frac{\gamma_1}{2} \sqrt{\frac{\gamma_1}{2\pi}} a_1 + \sqrt{\gamma_1} b_{in} \right) - \left(\frac{\gamma_1}{2} a_1^\dagger + \sqrt{\gamma_1} b_{in}^\dagger \right) [a_1, a_1] \sqrt{\frac{\gamma_1}{2\pi}}$$

$$c := a_1 \sqrt{\frac{\gamma_1}{2\pi}} \quad b_1 = b_{in}$$

$$= -i\omega_0 a_1 - \frac{\gamma_1^2}{4\pi} a_1 + \frac{\gamma_1}{\sqrt{2\pi}} b_{in}$$

$$\dot{a}_2 = -i\omega_0 a_2 - \frac{\gamma_2^2}{4\pi} a_2 + \frac{\gamma_2}{\sqrt{2\pi}} (b_{in} + \sqrt{\gamma_1} a_1)$$

$$c := a_2 \sqrt{\frac{\gamma_2}{2\pi}}$$

$$b_2 = b_{in} + \sqrt{\gamma_1} a_1$$

since z_1 and z_2 are the interaction points of these two terms

$$[O_S(t), c(t)] = e^{-iH_S t} [O_S(0), c(0)] e^{iH_S t} \xrightarrow{\text{combined op. on both } \mathcal{H}_S \text{ and } \mathcal{H}_B} [O_S(0) \otimes \mathbb{1}_B, c(0) \otimes \mathbb{1}_B]$$

→

$$b) \quad b(z > z_2, t) = b_{in}(t) + \sqrt{\gamma_1} a_1 + \sqrt{\gamma_2} a_2$$